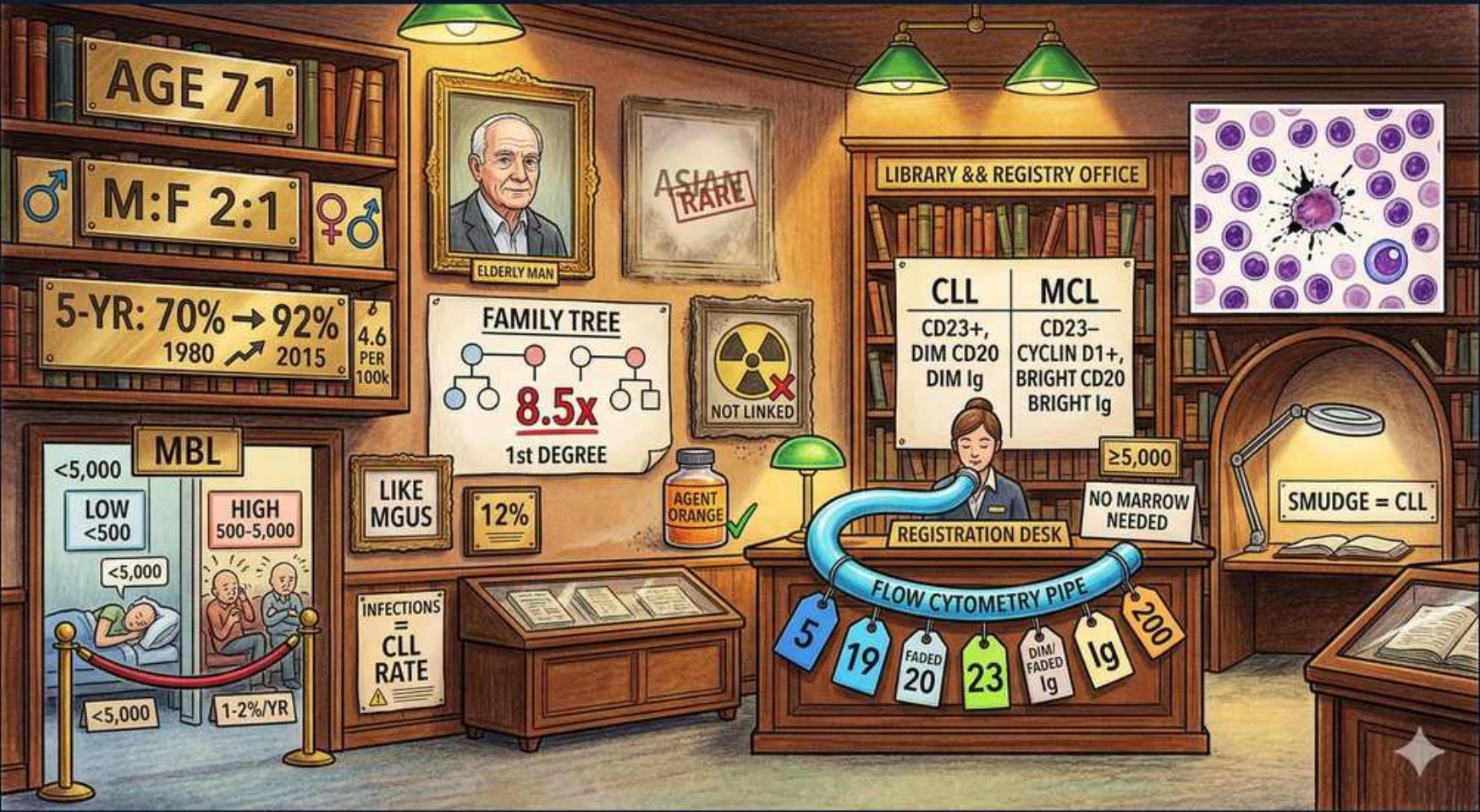


CLL — Epidemiology, Immunophenotype & Diagnosis



1. Age 71

WHAT YOU SEE

Wooden sign reading AGE 71

WHAT IT ENCODES

CLL is a disease of older adults — median age at diagnosis is ~70-72. It is the most common adult leukemia in the Western world, accounting for ~25-30% of all leukemias. Incidence rises steeply with age, and it is rare before age 40.

BOARD TRAP

Not a childhood leukemia — ALL is the pediatric one. A young patient with mature B-lymphocytosis should prompt a search for an alternative (reactive, MCL, or splenic marginal zone lymphoma) before settling on CLL.

2. M:F 2:1

WHAT YOU SEE

Sign showing male/female symbols, 2:1

WHAT IT ENCODES

Male predominance, roughly 1.5-2:1. CLL is also notably more common in people of European/Western descent and uncommon in East Asian populations, a difference that persists with migration and points to genetic rather than environmental drivers.

3. 5-yr 70→92%

WHAT YOU SEE

Survival sign 70%→92%, 1980 vs 2015

WHAT IT ENCODES

Indolent disease with a generally long natural history; 5-year overall survival has risen from ~70% to >90% with the era of targeted agents (BTK and BCL2 inhibitors). Many patients die WITH rather than FROM their CLL.

BOARD TRAP

Despite improved survival, CLL is generally considered incurable with standard therapy (FCR is the only regimen producing a functional-cure plateau, and only in IGHV-mutated patients).

4. 1st-degree 8.5x

WHAT YOU SEE

Family tree with 8.5x, 1st degree

WHAT IT ENCODES

The strongest known risk factor is family history — first-degree relatives of CLL patients carry roughly an 7-8.5x increased risk, the highest familial clustering of any leukemia. MBL is also more prevalent in these relatives.

BOARD TRAP

Agent Orange / herbicide exposure has been epidemiologically associated, but the classic leukemogen — ionizing radiation — is NOT linked to CLL (see the 'not linked' radiation sign).

5. Not linked

WHAT YOU SEE

Radiation symbol with NOT LINKED

WHAT IT ENCODES

Uniquely among the leukemias, CLL is NOT associated with ionizing radiation exposure (unlike AML, CML, and ALL, which all rise after radiation). It is also not caused by prior chemotherapy or benzene the way myeloid leukemias are.

BOARD TRAP

Radiation is a classic distractor on exams — if the stem emphasizes radiation/atomic-bomb/chemo exposure, think AML or CML, not CLL.

6. MBL <5,000

WHAT YOU SEE

MBL box, clonal B-cells <5,000

WHAT IT ENCODES

Monoclonal B-cell lymphocytosis (MBL): a clonal CLL-phenotype B-cell population <5×10⁶/L (<5,000/μL) in blood with NO lymphadenopathy, organomegaly, cytopenias, or disease symptoms. It is an obligate precursor — essentially all CLL passes through an MBL phase.

BOARD TRAP

The single number that separates MBL from CLL is the count: ≥5,000/μL monoclonal B-cells (≥5×10⁶/L) sustained = CLL. Below that with no other features = MBL, not CLL.

7. MGUS-like 1-2%/yr

WHAT YOU SEE

LIKE MGUS sign, 12% / 1-2%/yr

WHAT IT ENCODES

High-count MBL progresses to overt CLL requiring treatment at ~1-2% per year — directly analogous to MGUS→multiple myeloma. Low-count MBL almost never progresses. MBL is found in several percent of healthy adults over 60.

BOARD TRAP

The 1-2%/yr progression rate is the MGUS parallel; do not confuse it with the much higher progression of smoldering myeloma (~10%/yr).

8. CLL ≥5,000

WHAT YOU SEE

Library card CLL, CD23+ dim CD20 dim Ig

WHAT IT ENCODES

CLL immunophenotype: CD5+, CD19+, CD20 (dim), CD23+, with dim/weak surface immunoglobulin (light-chain restricted κ or λ). The combination of a T-cell marker (CD5) co-expressed on a clonal B-cell (CD19/CD20) with CD23 positivity and dim sIg is the diagnostic signature.

BOARD TRAP

Mantle cell lymphoma is the great mimic: also CD5+ but CD23-NEGATIVE, cyclin D1+ (t(11;14)), and bright surface Ig (see MCL card). Missing this leads to under-treating an aggressive lymphoma as indolent CLL.

9. MCL contrast

WHAT YOU SEE

MCL card: CD23-, cyclin D1+, bright Ig

WHAT IT ENCODES

Mantle cell lymphoma (MCL) shares CD5+ with CLL but is distinguished by CD23-negative, cyclin D1+ / SOX11+, t(11;14)(q13;q14) translocation, and bright sIg. MCL is far more aggressive and requires intensive therapy.

BOARD TRAP

CD5+ alone does NOT equal CLL. The discriminator is CD23: positive in CLL, negative in MCL. If CD5+/CD23- → check cyclin D1 and treat as MCL.

10. Flow cytometry

WHAT YOU SEE

FLOW CYTOMETRY PIPE / registration desk

WHAT IT ENCODES

Diagnosis is made by peripheral blood flow cytometry demonstrating $\geq 5 \times 10^6$ /L clonal B cells with the CLL phenotype — no tissue is required. The Matutes/CLL flow score (CD5, CD23, dim CD20, dim sIg, CD79b/FMC7 negative) formalizes the panel.

BOARD TRAP

A bone marrow biopsy is NOT required to diagnose CLL and is not part of the diagnostic criteria (see 'no marrow needed').

11. No marrow needed

WHAT YOU SEE

Sign NO MARROW NEEDED

WHAT IT ENCODES

CLL is diagnosed on peripheral blood flow cytometry alone. Bone marrow examination is reserved for clarifying the cause of cytopenias (marrow infiltration vs autoimmune vs treatment effect), not for diagnosis.

BOARD TRAP

If a question diagnoses CLL only after marrow biopsy, it is testing whether you know blood flow is sufficient — marrow is unnecessary.

12. Smudge = CLL

WHAT YOU SEE

SMUDGE = CLL lamp sign

WHAT IT ENCODES

Smudge (smear / basket) cells on the peripheral blood film are the classic morphologic clue of CLL. Their number correlates inversely with disease aggressiveness — more smudge cells tend to associate with more indolent (mutated IGHV) disease.

13. Smudge cells

WHAT YOU SEE

Purple smear with broken/smudged lymphocytes

WHAT IT ENCODES

The malignant CLL lymphocytes are fragile and rupture during slide preparation, producing smudge/smear cells. The intact cells are small, mature lymphocytes with clumped 'soccer-ball' chromatin and scant cytoplasm.

BOARD TRAP

Smudge cells are an artifact of fragility, not a true cell type; adding albumin to the smear reduces them — useful to know they are CLL-associated but not pathognomonic in isolation.

14. Infections ↑ rate

WHAT YOU SEE

INFECTIONS ↑ CLL RATE warning

WHAT IT ENCODES

Recurrent infections from progressive hypogammaglobulinemia (impaired humoral immunity) are the leading cause of morbidity and mortality in CLL. Encapsulated organisms and reactivations are common, and infection risk is compounded by treatment.

BOARD TRAP

Hypogammaglobulinemia drives infections; IVIG is given for recurrent serious bacterial infections, NOT routinely. CLL also predisposes to autoimmune cytopenias (AIHA, ITP) — a Coombs-positive anemia in CLL is immune, not marrow failure.

15. Autoimmune cytopenias

WHAT YOU SEE

No standalone drawing — covered by the family/immune-complication note panel on the left; flags CLL's tendency to immune cytopenias

WHAT IT ENCODES

CLL is the classic cause of secondary autoimmune hemolytic anemia (warm IgG, Coombs/DAT positive) and immune thrombocytopenia. These are immune-mediated and respond to steroids/rituximab — distinct from cytopenias due to marrow packing.

BOARD TRAP

A Coombs-positive anemia in a CLL patient is AIHA (immune) and does NOT by itself upstage to Rai III — staging cytopenias must be from marrow infiltration, not autoimmune destruction.